Stack Overflow Survey Data

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OUTLINE

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EXECUTIVE SUMMARY

Summaries

- The workforce is dominated by young professionals

The largest age group is **25–34 (41.3%)**

- Majority hold Bachelor's degrees
- Strong presence of Master's degree holders
- Represents early-to-mid career professionals

- Python and JavaScript are the most in-demand programming languages
- PostgreSQL is the leading database for both current use and future demand
- Cloud platforms are dominated by AWS and Microsoft Azure

INTRODUCTION

Purpose of the report, Target Audience & Value of the analysis

- To analyze global developer survey data and identify key technology trends
- To compare current technology usage with future demand
- Target audience: decision-makers, hiring managers and technical teams
- Provides data driven insights for workforce planning and technology adoption
  - Supports informed hiring and upskilling decisions
  - Helps organizations align with emerging industry trends

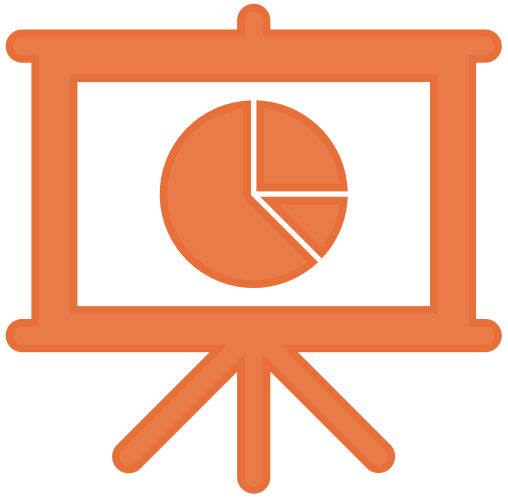
METHODOLOGY

Data sources, Data collection & Data wrangling steps

- Data source: Developer survey dataset (Stack Overflow survey)
- Data collection: Extracted relevant variables for analysis (e.g., languages, databases, age, compensation)
- Data wrangling: Cleaned and prepared data for analysis
  - Handled missing values and removed inconsistencies
  - Converted data types and standardized categories

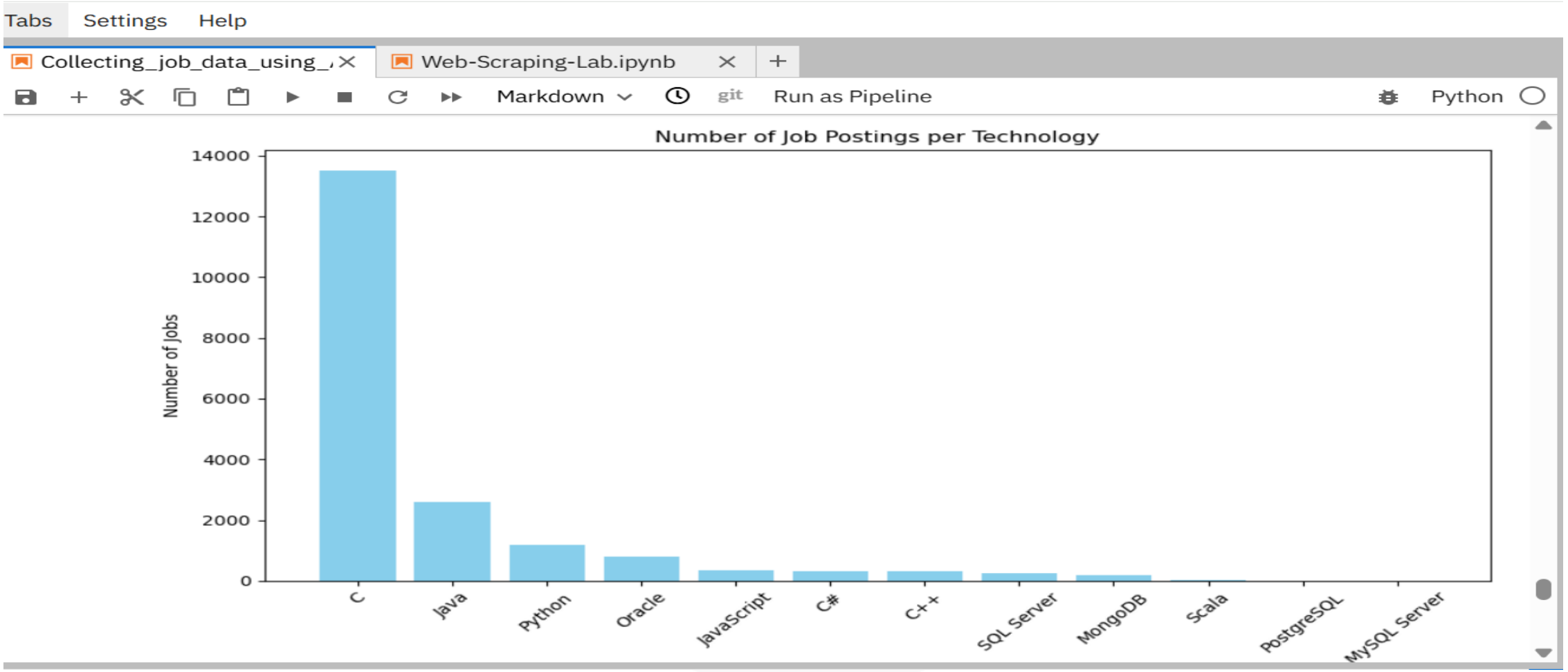
  Data analysis: Aggregated and visualized data using charts and dashboards

Results

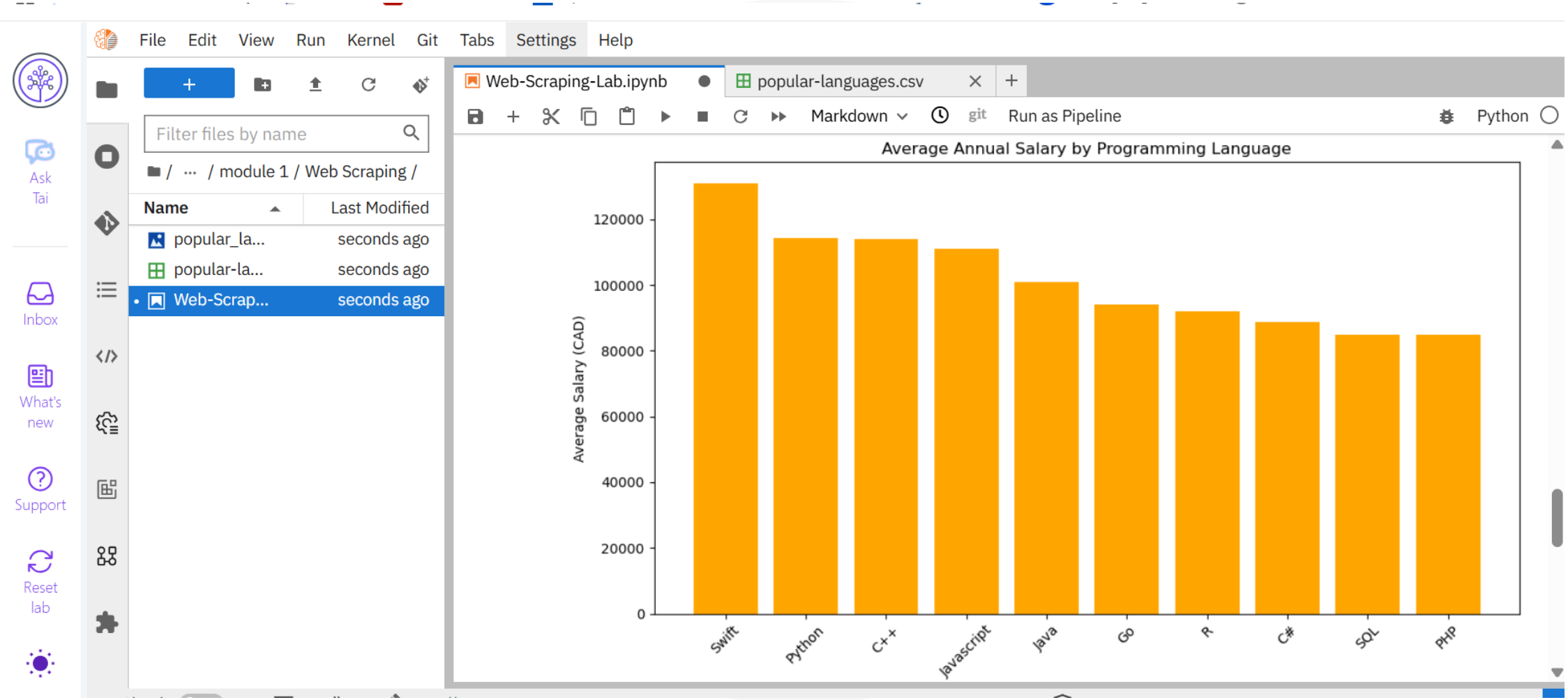


Results of the analysis are presented in the next slides.

JOB POSTINGS



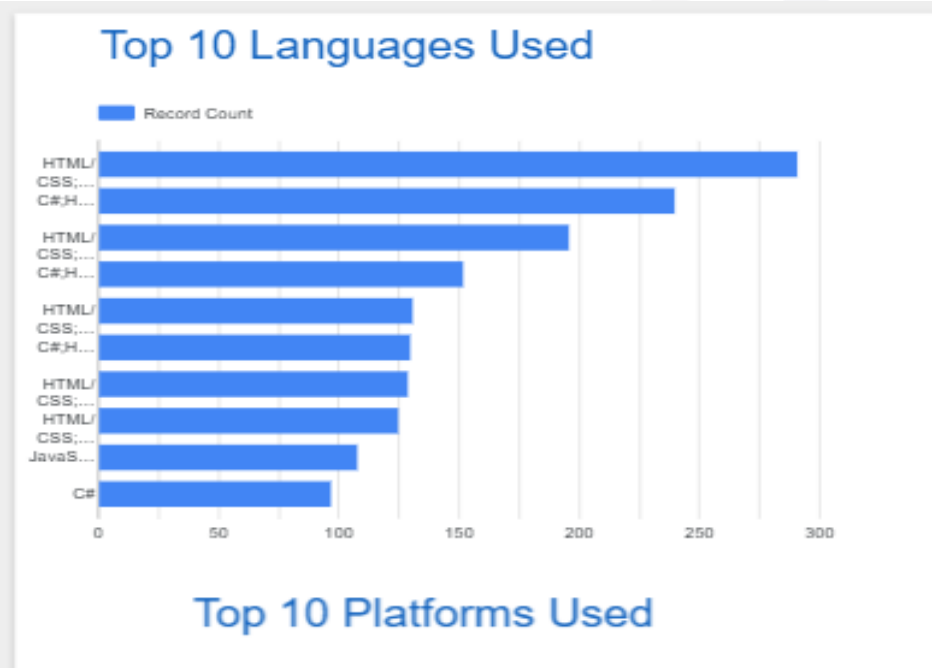
POPULAR LANGUAGES



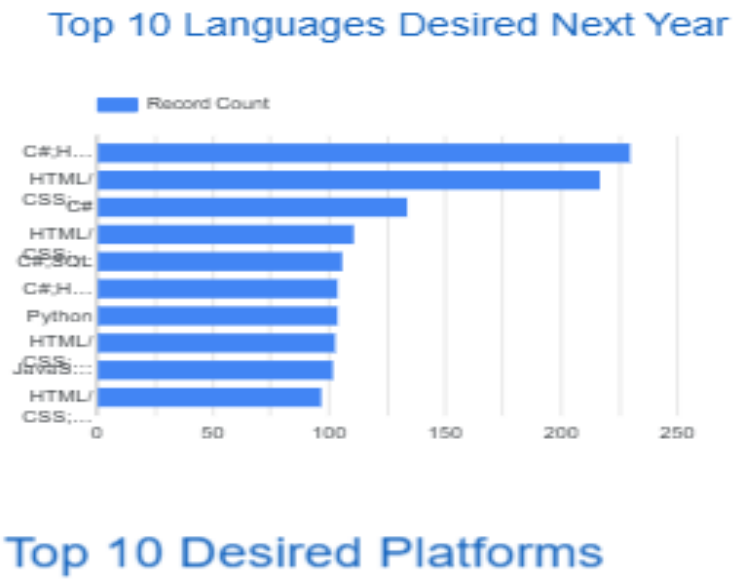
PROGRAMMING LANGUAGE TRENDS

Summaries of key trends shown in Programming Languages.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- HTML/CSS, SQL and TypeScript are very popular among developers
- C# is the most desired language for next year
- Legacy languages like C, C++ and PHP show a decline in future interest

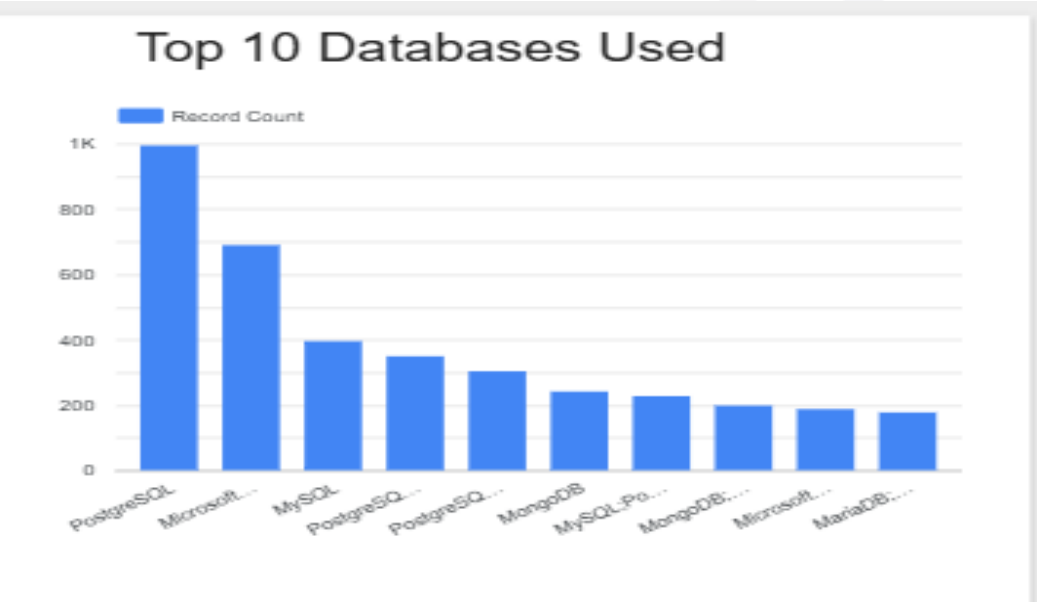
Implications

- Organizations should prioritize hiring for full-stack and data integrated web development skills
- Developers need to upskill in C# to remain competitive
- Training programs should gradually phase out legacy languages in favor of modern, in-demand technologies

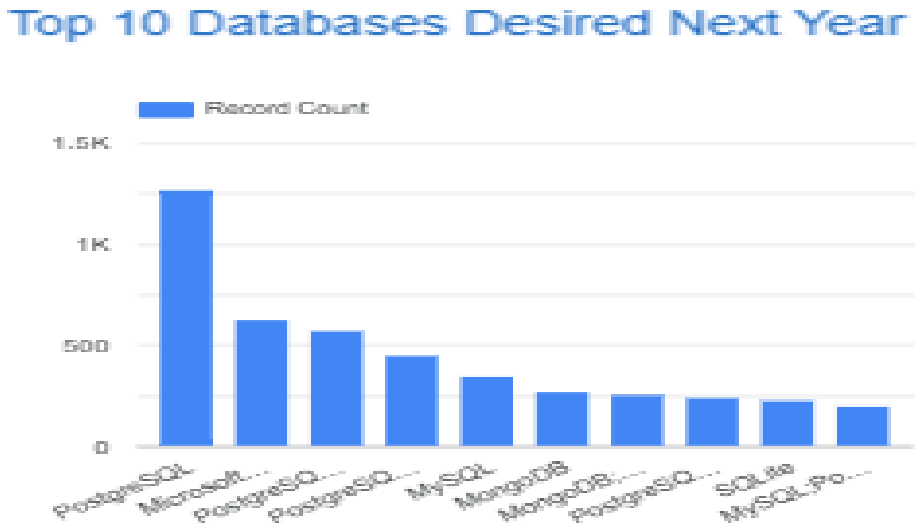
DATABASE TRENDS

Summaries of key trends shown in Programming Languages.

Current Year



Next Year



Top 10 Desired Web Frameworks

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the leading database for both current use and future demand
- MongoDB shows strong growth, highlighting increased interest in NoSQL
- Microsoft SQL Server and MySQL are declining in future preference

Implications

- Organizations should standardize on PostgreSQL to align with market trends
- Investment should go into MongoDB skills to support emerging NoSQL adoption
- Reduce reliance on legacy, high-cost database systems to optimize infrastructure and recruitment

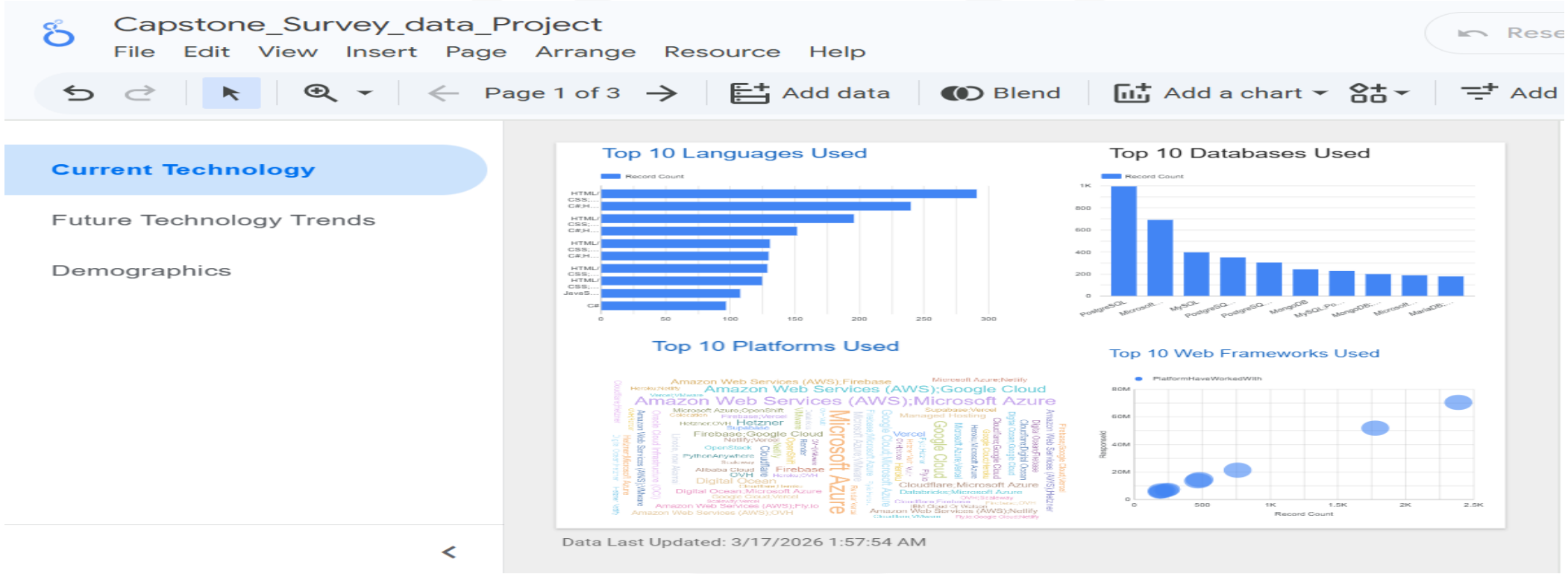
DASHBOARD

The next slides present the dashboards highlighting current and future technology trends.

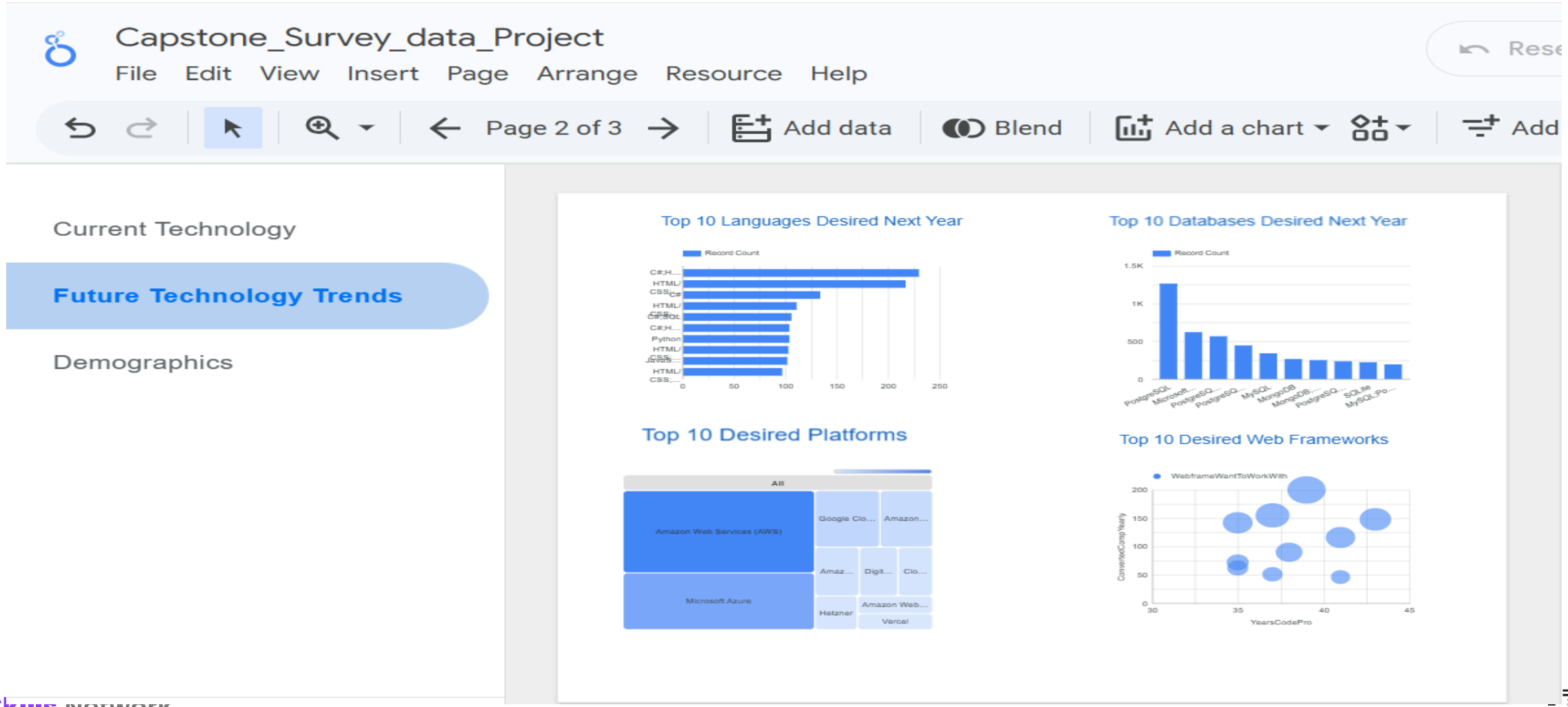
DASHBOARD TAB 1: Current Technology Usage

Dashboard Insight

- AWS and Microsoft Azure dominate the cloud infrastructure landscape



DASHBOARD TAB 2: Future Technology Trends

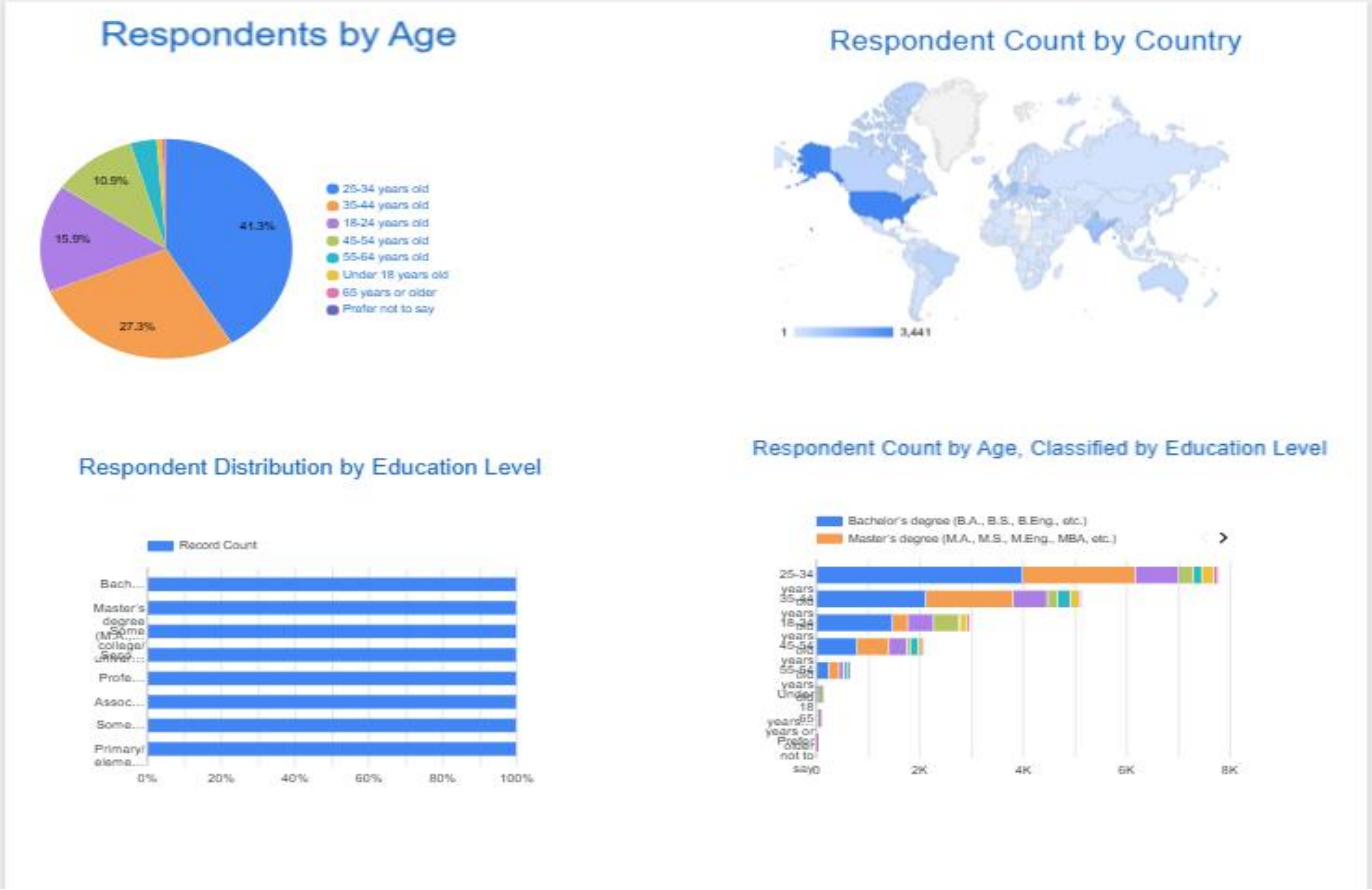


DASHBOARD TAB 3: Demographics

Current Technology

Future Technology Trends

Demographics



DISCUSSION

The developer workforce is largely composed of young, educated professionals

There is a strong shift toward modern programming languages such as Python, JavaScript and C#

Database trends show a clear move toward open-source and NoSQL solutions

Cloud computing remains central, with AWS and Microsoft Azure leading adoption

OVERALL FINDINGS & IMPLICATIONS

Findings

- The workforce is dominated by the 25–34 age group (41.3%)
- PostgreSQL is the leading database for both current and future use
- Cloud infrastructure is concentrated in AWS and Microsoft Azure

Implications

- Organizations should focus hiring on mid-career, highly educated professionals
- Adopt modern, open source technologies such as PostgreSQL and cloud platforms
- Invest in continuous upskilling in high demand areas like cloud computing and full stack development

CONCLUSION

- The technology landscape is rapidly evolving toward modern, scalable solutions
- Young professionals are driving adoption of new tools and platforms
- Organizations must align hiring and technology strategies with current trends
- Continuous learning and adaptation are critical to remain competitive

APPENDIX

Top 10 Platforms Used
Amazon Web Services (AWS);Firebase
Microsoft Azure;Netlify
Amazon Web Services (AWS);Google Cloud
Amazon Web Services (AWS);Microsoft Azure
Microsoft Azure;OpenShift
Firebase;Vercel
Hetzner;OVH
Hetzner
Supabase
Firebase;Google Cloud
Netlify;Vercel
OpenStack
PythonAnywhere
Cloudflare
Alibaba Cloud
OVH
Firebase
Digital Ocean
Digital Ocean;Microsoft Azure
Amazon Web Services (AWS);Fly.io
Amazon Web Services (AWS);OVH
Microsoft Azure
Supabase;Vercel
Managed Hosting
Google Cloud
Vercel
Cloudflare;Microsoft Azure
Databricks;Microsoft Azure
Amazon Web Services (AWS);Netlify
Amazon Web Services (AWS);Hetzner
Digital Ocean;Firebase
Cloudflare;Digital Ocean
Cloudflare;Google Cloud
Microsoft Azure;Vercel
Heroku;Microsoft Azure